

Stem Cells, Kidney Regeneration, and Gene and Cell Therapy in Nephrology

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KEY POINTS

- Effective approaches to arrest and reverse progressive chronic kidney disease are limited to several specific etiologies, and restoration of lost nephron function remains to be reliably achieved.
- Preclinical studies have demonstrated promising results in the treatment of chronic kidney disease and acute kidney injury with stem cell–based therapy of various origins.
- Clinical trials have demonstrated the safety and tolerance of short-term therapy using stem cells in patients with chronic kidney disease and acute kidney injury.
- The efficacy of stem cell–based therapy in patients with kidney disease warrants further large-scale and long-term clinical studies.
- Stem cells and their exosomes demonstrate potential in kidney regeneration, but there remain challenges for clinical use.
- Cell- and gene-based therapies are likely to complement current therapeutics.

Chronic kidney disease (CKD) is a worldwide public health problem, with adverse outcomes of kidney failure, cardiovascular disease, and premature death. Approximately 10% of the population worldwide is affected by CKD. The prevalence of CKD has been increasing at an alarming rate over the past 2 decades because of the aging population and growing prevalence of diabetes, hypertension, and obesity, exacerbated by the frequent occurrence of superimposed acute kidney injury (AKI). A forecasting analysis suggests that the number of global years of life lost due to CKD will increase from ~26 million in 2016 to 52.5 million in 2040, and deaths will rise from 1.2 million in 2016 to 3.1 million in 2040.¹ These increases are predicted to move CKD in the years of life lost rankings from 16th in 2016 to 5th in 2040. However, current best clinical practices only slow the progress of renal functional decline in CKD. Kidney transplantation and dialysis are the only options

for the management of end-stage kidney failure (ESKF), which results in a significant burden on the health system. In addition, these treatment modalities are associated with significantly reduced quality of life and reduced lifespan. Recent developments in stem cell, regenerative medicine, cell, and gene therapies have heralded the potential for novel treatments for patients with CKD. Stem cell–based therapeutics have the potential to significantly improve lives of those living with CKD.

STEM CELLS

Stem cells are undifferentiated cells with high potential for self-renewal, proliferation, and differentiation. Stem cells exist in various embryonic and adult tissues such as bone marrow, fat, blood, amniotic fluid, and umbilical cord. Stem

cells can be further classified into subtypes as detailed in Table 84.1.

In 1981, Gail Martin² isolated ESCs from inner cell masses of late blastocysts cultured in medium conditioned by an established teratocarcinoma stem cell line, and Evans and Kaufman reported establishment in culture of pluripotent cells from mouse embryos.³ Thomson and colleagues⁴ and Reubinoff and colleagues⁵ reported embryonic stem cell lines derived from human blastocysts in 1998 and 2000, respectively.

Induced pluripotent stem cells (iPSCs) can be generated by reprogramming embryonic or somatic cells. iPSCs can also be induced chemically by small molecules. In 2006, Takahashi and Yamanaka⁴ successfully reprogrammed mouse embryonic fibroblasts into iPSCs by introducing four factors, Oct3/4, Sox2, c-Myc, and Klf4, under embryonic stem (ES) cell culture conditions, and these iPSCs exhibit the morphology and growth properties of ES cells and express ES cell marker genes. In the following year, Takahashi and colleagues⁵ generated iPSCs from adult human dermal fibroblasts with the same four factors: Oct3/4, Sox2, Klf4, and c-Myc, and these human iPSCs were similar to human ES cells in morphology, proliferation, surface antigens, gene expression, epigenetic status of pluripotent cell-specific genes, and telomerase activity. In 2013, Hou and colleagues generated induced pluripotent stem cells from mouse somatic cells using a combination of seven small-molecule compounds, and the chemically induced pluripotent stem cells resemble ES cells in terms of their gene expression profiles, epigenetic status, and potential for differentiation and germline transmission.⁶ In 2022, Guan and colleagues⁷ generated induced pluripotent stem cells by creating an intermediate plastic state of human somatic cells through chemical reprogramming. These stem cells exhibit key features of embryonic stem cells.⁷ Recently, the potential for iPSC technology-based regenerative medicine has been investigated for kidney diseases.^{8,9}

Mesenchymal stem cells (MSCs) have been most widely studied for their applications in human health disease. MSCs can be collected and isolated from various tissues including bone marrow, the umbilical cord, and fat. MSC-based therapy for kidney disease has been recently reported.^{10,11} Moreover, MSCs have been isolated from many other tissues. Neural stem cells (NSCs) are found in adult mammalian brains, and NSCs have the capacity to self-renew and undergo multipotential differentiation. Reynolds and Weiss reported that isolated cells from the adult mouse striatum have the capacity to divide and differentiate into neurons and astrocytes.¹² In 2006, Herrera and colleagues¹³ isolated and characterized a population of human liver stem cells (HLSCs). These cells expressed the MSC markers CD29, CD73, CD44, and CD90 but not the hematopoietic stem cell markers CD34, CD45, CD117, and CD133. Zhang and colleagues¹⁴ had identified a human hepatic stem/progenitor population in the CDCP1⁺CD90⁺CD66⁻ subpopulation in human primary fetal liver cells. Heart stem cells are multipotent cells residing in the adult mammalian heart that are capable of self-renewing and generating coronary vessels and cardiomyocytes. Heart stem cells can contribute to new cardiomyocyte formation following experimental myocardial infarction in mice.¹⁵ Hence human ESCs and iPSCs can be differentiated to adult stem cells.¹⁶

Table 84.1 Classification of Stem Cells

Stem Cells	Characterizations and Functions
Embryonic stem cells (ESCs)	Human ESCs can differentiate into any cell type (pluripotent) and potentially be used for various medical conditions including tissue injury. However, application is limited due to ethical reservations.
Adult stem cells (ASCs)	ASCs, also known as somatic stem cells, are undifferentiated cells existing in various tissues and can differentiate into different cell types. ASCs include hematopoietic stem cells and mesenchymal stem cells (MSCs).
Induced pluripotent stem cells (iPSCs)	iPSCs are generated by genetic reprogramming through introducing specific genes into adult cells to acquire characteristics of embryonic stem cells. iPSCs generated from patient's own cells do not cause immunologic reactions when used as autotherapy. Thus iPSCs have potential in personalized medicine and tissue repair.
Progenitor cells	These cells exhibit differentiation potential between stem cells and mature cells. These cells play a part in maintaining the wound healing process to replace the damaged or dead cells, participating in the normal cell turnover of organs.

Human embryonic stem cells (hESCs) can theoretically be used for cell therapies. However, the clinical application of hESCs leads to ethical concerns. Moreover, hESCs may induce immune responses in patients that limit their therapeutic potential. Human-induced pluripotent stem cells (hiPSCs) have significant advantages for autologous transplantation. Since there are no ethical concerns and no immune response is elicited, hiPSCs are a preferred source of stem cells for clinical application. However, concerns regarding the potential for tumorigenicity remain. Similar to hiPSCs, adult stem cells (ASCs) have no ethical concerns, but their proliferation and differentiation abilities are relatively limited. Many ongoing clinical trials are using stem cell therapies for multiple clinical conditions. These trials should address ongoing areas of uncertainty, which include defining the optimal sites for collection of stem cells, best areas for clinical application, number of cells required, route of administration, preservation of cells limiting the risks of malignancy, systemic immunosuppression, and maldifferentiation.

KIDNEY STEM CELLS

In the adult kidney, there are two different sources of stem/progenitor cells including resident renal stem cells and circulating stem cells.

KIDNEY-DERIVED STEM CELLS

Kidney cells with mesenchymal stem characteristics have been derived and propagated in vitro from the postnatal

kidney.¹⁷ Subsequently, resident MSCs have been isolated and characterized from human glomeruli.^{18,19} It was found that human glomeruli deprived of the Bowman capsule contain a population of CD133⁺CD146⁺ cells expressing MSC markers, as well as the renal stem cell markers CD24 and Pax-2. These renal stem cell markers are not found in bone marrow stem cells (BMSCs). These cells exhibited a capacity for self-renewal, clonogenicity, and multipotency. These cells also acquired mesangial cell markers such as α -smooth muscle actin (α -SMA) and angiotensin II (AT-II) receptor I when cultured in appropriate conditions. It was concluded that a multipotent mesenchymal cell population is resident in human glomeruli and may have a role in physiologic cell turnover and/or in response to glomerular injury.^{18,19} A further report demonstrated that CD133⁺CD24⁺ cells in the adult human kidney consist of a hierarchical population of progenitors that are arranged in a precise sequence within the Bowman capsule and exhibit heterogeneous potential for differentiation and regeneration. These cells improved chronic glomerular damage in a model of doxorubicin (Adriamycin)-induced nephropathy in mice.²⁰ CD133⁺CD24⁺CD106⁺ cells exhibited a high proliferative rate and could differentiate toward the podocyte, as well as the tubular lineage. CD133⁺CD24⁺CD106⁻ cells showed a lower proliferative capacity and displayed a committed phenotype toward the tubular lineage. Both these populations displayed the capacity to engraft within the kidney, generate new tubular cells, and improve renal function after injection into SCID mice with acute tubular injury.²¹ CD133⁺CD24⁺CD106⁻ cells proliferate upon tubular injury, becoming the predominant component of the regenerating epithelium in patients with acute or chronic tubular damage. Thus CD133⁺CD24⁺CD106⁻ cells represent tubular-committed progenitors that display resistance to apoptotic stimuli and exert regenerative potential for injured tubular tissue.²¹ Further studies have confirmed that CD24⁺CD133⁺ cells in the Bowman capsule contribute to tubular regeneration.²²⁻²⁴

The presence of stem cells in kidney tubules is well reported. In 2003, Maeshima and colleagues²⁵ identified renal progenitor-like tubular cells that participate in regeneration processes after kidney injury. These renal progenitor-like tubular cells actively proliferate and eventually differentiate into epithelial cells during tubular regeneration. Gupta and colleagues²⁶ reported a population of cells with characteristics of renal stem cells isolated from rat kidneys. The features of these cells include spindle-shaped morphology; self-renewal for >200 population doublings without evidence for senescence; normal karyotype and DNA analysis; and expression of vimentin, CD90 (thy1.1), Pax-2, and Oct4 but not cytokeratin, MHC class I or II, or other markers of more differentiated cells.²⁶ These cells can differentiate into renal tubules when injected under the capsule of an uninjured kidney or intraarterially after renal ischemia-reperfusion injury (IPI). Challen and colleagues²⁷ reported that adult kidney side population (SP) cells demonstrated multilineage differentiation *in vitro*, whereas microinjection into mouse metanephroi showed that SP cells had a 3.5- to 13-fold greater potential to contribute to developing kidneys than the non-SP main population cells. Many studies have shown that CD24⁺CD133⁺ stem/

progenitor cells in the tubules can regenerate tubular cells and improve renal function after kidney injury.^{21,24,28,29} Lazzeri and colleagues³⁰ identified and characterized renal multipotent progenitors in human embryonic kidneys that share CD24 and CD133 surface expression with adult renal progenitors. These cells have the capacity for self-renewal and multilineage differentiation, regenerate different portions of the nephron, reduce tissue necrosis and fibrosis, and significantly improve renal function after injection into SCID mice with AKI.³⁰

Kidney resident MSCs in adult kidneys have been isolated, and their characteristics are like those of ESCs.³¹⁻³³ These cells can differentiate into various lineages including mesodermal lineages, endothelial cells (ECs), and erythropoietin-producing fibroblasts. These cells promote improvement of renal function following kidney injury. As reviewed by Bruno and colleagues,³⁴ an MSC-like population has been demonstrated when kidney injury occurs. These cells are considered to provide support for kidney regeneration by recapitulating the role they have in embryonic life.³⁴

BONE MARROW-DERIVED STEM CELLS

It is well accepted that bone marrow-derived stem/progenitor cells (BMSCs) can be released into the peripheral blood and then distributed in all organs and tissues of the body. Patschan and colleagues³⁵ confirmed endothelial progenitor cell (EPC) mobilization and homing during acute renal ischemia. Hematopoietic stem cells contribute to the regeneration of renal tubules after renal IRI in mice.³⁶ It is reported that EPCs are mobilized into the damaged glomeruli and are likely to be involved in glomerular capillary repair in damaged glomeruli in experimental glomerulonephritis in mice.³⁷ Human CD34⁽⁺⁾ hematopoietic stem/progenitor cells (HSPCs) have been demonstrated to promote kidney repair and regeneration by paracrine mechanisms, rather than replacement of vasculature in an IRI model in mice.³⁸ By studying the time course of EPC and HSC stimulation and homing following induction of selective EC injury in the mouse kidney, along with various angiogenic factors potentially involved in EC repair and progenitor cell stimulation, it was found that selective EC injury leads to a marked stimulation of EPCs, HSCs, and various angiogenic factors that orchestrate microvascular repair.³⁹

BMSCs have been identified in both histologically normal mouse kidneys and human transplanted kidneys with damage from a variety of causes, which suggested that bone marrow cells contribute to both normal turnover of renal epithelia and regeneration after damage.⁴⁰ It is reported that bone marrow-derived cells may have the potential to differentiate into glomerular mesangial cells.⁴¹ A report showed that recruitment of bone marrow-derived cells into glomeruli was dramatically facilitated in response to mesangiolysis evoked in anti-Thy1 antibody-mediated glomerulonephritis. In the regenerated glomeruli, 11% to 12% of glomerular cells were derived from the transplanted bone marrow and bone marrow-derived Thy1(+) cells kept increasing in number until the remodeling ceased, finally making up 7% to 8% of glomerular cells. These results indicate the contribution of bone marrow-derived cells to repopulating mesangial cells during glomerular remodeling *in vivo*.⁴²

BMSCs can be induced to differentiate along podocytic lines *in vitro*. Thus BMSCs may be useful in the development of stem cell–based reconstitution of glomeruli that are damaged by disease.⁴³ As well reviewed by Imai and Ito,⁴⁴ bone marrow–derived cells have the capacity to differentiate into renal cells including mesangial cells, ECs podocytes, and tubular cells in the kidney, although controversy exists.

KIDNEY REGENERATION

The holy grail of renal therapeutics is to promote renal regeneration, rather than pathologic scar tissue in the form of tubular fibrosis and glomerulosclerosis. As reviewed by Tsuji and colleagues,⁴⁵ there are two different strategies to promote renal regeneration: kidney repair or *de novo* whole kidney development.

STEM CELL–BASED THERAPY IN ACUTE AND CHRONIC KIDNEY DISEASE

Stem cell–based therapy in kidney disease was described briefly earlier. Here the tissue-repairing function of stem cells in acute and chronic kidney injury is reviewed.

INDUCED PLURIPOTENT STEM CELLS IN KIDNEY REPAIR

An efficient system of homologous recombination of human pluripotent stem cells has been established to generate human intermediate mesoderm (IM) (an embryonic germ layer that gives rise to kidneys), which has the ability to differentiate into multiple cell types that constitute adult kidneys, and to form renal tubule–like structures.⁴⁶ An alternative system for differentiating hESCs and hiPSCs into cells expressing markers of the IM has been established through fibroblast growth factor signaling to induce IM differentiation in hPSCs, with features characteristic of kidney lineage cells.⁴⁷ Three-dimensional (3D) structures of the kidney including glomeruli with podocytes, as well as renal tubules with proximal and distal regions and clear lumina, have been successfully developed *in vitro* from PSCs by reevaluating the developmental origins of metanephric progenitors.⁴⁸ It is reported that the renal progenitors generated from hiPSCs are capable of reconstituting 3D proximal renal tubule–like structures *in vitro* and *in vivo* and have the potential to ameliorate AKI (AKI).⁴⁹ Transplantation of iPSCs without c-Myc reduced the expression of oxidative substances, proinflammatory cytokines, and apoptotic factors in kidney tissues subjected to IRI and eventually improved survival in rats subjected to ischemic AKI.⁵⁰ iPSC-derived conditioned medium has also been reported to attenuate AKI by downregulating the oxidative stress–related pathway following ischemia–reperfusion injury in rats.⁵¹

ENDOTHELIAL PROGENITOR CELLS IN KIDNEY REPAIR

EPC-based therapies have been studied in different models of AKI and CKD. Injection of renal artery–derived vascular progenitor cells (RAPCs) is associated with decreased serum creatinine after ischemia/reperfusion, reduced albuminuria, and improved blood flow in both animal models of acute ischemic and hyperperfusion injuries, with a small population of RAPC being shown to integrate with the renal microvasculature in the kidney. *In vitro* studies have also

shown that RAPCs can promote local endothelial migration in co-culture.⁵² Systemically injected early endothelial progenitor cells (eEPCs) reduced serum creatinine levels in C57/BL6N mice subjected to unilateral renal ischemia post uninephrectomy, significantly diminished interstitial fibrosis and prevented the loss of peritubular capillaries.⁵³ The effect of EPC treatment on the progression of CKD was determined in C57BL/6 mice with 5/6 nephrectomy–induced CKD. Renal function deteriorated steadily over time in control mice, whereas the EPC-treated CKD group showed less deterioration of renal function and reduced proteinuria, along with relatively preserved kidney structure. The expression of proinflammatory cytokines and adhesion molecules was decreased in the kidneys of the EPC-CKD group. These results indicate that EPCs trafficked into the injured kidney protected the kidney from inflammation and consequently resulted in functional and structural renal preservation.⁵⁴ Additional studies have shown that peripheral blood-derived endothelial progenitor cell therapy effectively inhibits the propagation of CKD and deterioration of renal function through enhancement of angiogenesis, blood flow, and antioxidative capacity, as well as suppression of inflammation, oxidative stress, apoptosis, and fibrosis in kidneys of rats subjected to 5/6 nephrectomy.⁵⁵

MESENCHYMAL STEM CELLS IN KIDNEY REPAIR

The renoprotective potential of pluripotent and adult stem cell therapy in experimental models of acute and chronic kidney injury has been intensively studied. MSCs are clearly shown to be the most efficient cells to initiate regeneration in damaged kidneys.⁵⁶ AKI-CKD transition is one of the major contributors to CKD. Yet there is still no reliable therapeutic approach for treatment of the AKI-CKD transition. MSCs have the potential to at least partially block AKI-CKD transition.⁵⁷ Treatment with MSC resulted in reduced inflammatory cytokines synthesis and a significant increase in the synthesis of adenosine triphosphate and number of renal cells undergoing mitosis (26%) in *ex vivo* kidneys (human kidney allografts from donation after cardiac death).⁵⁸ MSCs can differentiate into multiple kidney cells including epithelial, mesangial, and ECs. For example, MSC GFP-positive cells, injected intravenously, homed to the kidney of mice with glycerol-induced AKI but not the kidney of normal mice. These MSCs localized to the tubular epithelial lining and expressed cytokeratin, differentiated into tubular epithelial cells, and promoted morphologic and functional recovery. Moreover, these MSCs enhanced tubular proliferation.⁵⁹ hMSCs transdifferentiate to renal tubule epithelium after injection into the kidneys of newborn mice.⁶⁰ Gene transcripts during the differentiation of MSC into mesangial cells have been identified through systematic transcriptomic profiling.⁶¹ MSCs have also been successfully induced to differentiate into endothelial lineage cells with capacity to develop capillary networks and progressively form vessel-like structures.⁶² Hence MSCs can differentiate into multiple renal cells to effect renal repair and regeneration.

BONE MARROW–DERIVED MESENCHYMAL STEM CELLS IN KIDNEY REPAIR

In vivo infusion of MSCs protected mice from AKI and subsequent renal function impairment and severe tubular injury

in a model generated by glycerol-induced rhabdomyolysis, which was accompanied by a time-dependent increase in CD206-positive M2 macrophage infiltration.⁶³ A further study demonstrated that treatment with synthetic erythropoietin-pretreated BM-MSCs resulted in significantly reduced levels of serum IL-1 β and TNF- α and significantly increased levels of IL-10 in rats following ischemia/reperfusion-induced AKI, compared with control rats infused with untreated BM-MSCs. Synthetic erythropoietin-BM-MSCs improved renal function as indicated by reduced serum creatinine, blood urea nitrogen, and improved pathologic scores.⁶⁴ It was also reported that infusion of MSCs enhanced recovery of renal function in I/R-induced AKI in rats. MSCs were located in the kidney cortex after injection, and histologically, MSCs were predominantly located in glomerular capillaries.⁶⁵ MSC therapy has also been shown to promote renal repair by limiting glomerular podocyte and progenitor cell dysfunction in experimental Adriamycin-induced nephropathy.⁶⁶ It has been shown that BM-MSCs can differentiate into mesangial cells *in vivo* in athymic mice with glomerular injury and accelerate glomerular healing in experimental glomerulonephritis.^{67,68} In addition to supporting glomerular repair, human BM-MSCs have been shown to decrease proximal tubular epithelial cell injury and ameliorate renal dysfunction, resulting in reduced mortality in a mouse model of cisplatin-induced AKI. BM-MSCs were predominantly localized in peritubular areas and shown to reduce renal cell apoptosis and increase proliferation. BM-MSCs protect against AKI-related peritubular capillary pathology, including EC abnormalities, such as cytoplasmic swelling, marked reduction in volume density of ECs and capillary lumen, leukocyte infiltration, and low EC and lumen volume density.⁶⁹ The effect of MSC treatment for CKD has been reviewed elsewhere.^{10,11}

ADIPOSE TISSUE-DERIVED MESENCHYMAL STEM CELLS IN KIDNEY REPAIR

Infusion of AD-MSCs has been reported to ameliorate renal dysfunction and kidney injury in a rat model of AKI induced by cisplatin, with a parallel increased survival in treated rats. AD-MSCs reduced apoptotic cell death, activation of p53, c-Jun N-terminal Kinase (JNK), extracellular signal-regulated kinase, and the expression of inflammation-related molecules in the injured kidney. These results show that human AD-MSCs exert a paracrine-protective effect on cisplatin nephrotoxicity at multiple target sites.⁷⁰ It was also reported that human adipose tissue-derived stromal cells (hASCs) cultured with low (2%) serum showed a greater potential for tissue regeneration in folic acid-induced AKI in rats through paracrine effects, with hepatocyte growth factor (HGF) being one of the key mediators.⁷¹ hAD-MSCs have been shown to reduce oxidative stress markers and have a renoprotective effect against cisplatin-induced AKI in Sprague-Dawley rats.⁷² These studies raise the possibility that easily obtained autologous AD-MSCs could be used when AKI is expected to occur, such as in patients undergoing chemotherapy with platinum-based therapies.

UMBILICAL CORD-DERIVED STEM CELLS UMBILICAL CORD-MESENCHYMAL STEM CELLS IN KIDNEY REPAIR

An isolation protocol has been developed to obtain MSCs from the umbilical cord matrix. The isolated cells expressing all core markers for MSCs are clonogenic,

retain long telomeres, can undergo several population doublings *in vitro*, and can be differentiated into mature mesenchyme, such as bone and adipose tissue.⁷³ UC-MSCs are reported to be more potent immunosuppressors and less immunogenic than BM-MSCs, which may be due to increased levels of immunomodulatory surface proteins, such as CD200, CD273, CD274, and cytokines like IL1 β , IL-8, LIF, and TGF β 2.⁷⁴ UC-MSCs have a greater proliferative capacity compared with AD-MSCs and BM-MSCs.^{75,76} UC-MSCs have been shown to traffic into the postischemic kidney and attenuate renal functional decline and tubular injury in mouse models of IRI. UC-MSCs reduced expression of interferon- γ but enhanced expression of vascular endothelial growth factor (VEGF) in the injured kidney.⁷⁷ UC-MSCs after IRI in male rats improved glomerular filtration (decreased plasma urea and creatinine and increased creatinine clearance) and histologically had higher expression of aquaporin 2 and less macrophage infiltration in the kidneys. The IRI-induced pro-oxidative and senescence-related proteins and microRNAs and reduction in Klotho expression were reversed in the kidney after IRI by UC-MSC treatment.⁷⁸ Further studies have confirmed the renoprotective role of UC-MSCs in animal models of CKD.^{79,80} The number of UC-MSCs recruited to the injured kidneys was increased significantly in a rat model of streptozotocin-induced diabetic kidney disease (DKD), and UC-MSC treatment ameliorated the increase in 24-hour urinary protein, serum creatinine, urea nitrogen, and renal hypertrophy index and significantly reduced renal vacuole degeneration, inflammatory cell infiltration, and renal interstitial fibrosis. *In vitro* experiments, UC-MSC conditioned medium and UC-MSC-derived exosomes decreased the production of proinflammatory cytokines (IL-6, IL-1 β , and TNF- α) and profibrotic factor (TGF- β) in renal tubular epithelial cells and glomerular ECs exposed to high glucose. Moreover, UC-MSCs secrete large amounts of growth factors considered to be renoprotective including EGF, FGF, HGF, and VEGF. These results suggested UC-MSCs can effectively improve the renal function and inhibit inflammation and fibrosis in an animal model of diabetes-induced CKD.⁸⁰ The low immunogenicity and immunoregulatory activity of UC-MSCs make them particularly attractive for therapeutic exploitation. However, the current method to isolate UC-MSCs results in no specific phenotype and the isolation and culture should be improved and standardized. The malignant tendency and malignant transformation after long-term cultivation of UC-MSCs also need further investigation.

STEM CELL-DERIVED EXOSOME-BASED THERAPY IN KIDNEY DISEASE

Extracellular vesicles (EVs) are secreted membranous structures including exosomes and microvesicles. Microvesicles (50–1000 nm in diameter) comprise a large and heterogeneous group of vesicles with different membranes depending on their origin and morphology. MSC-released exosomes (30–150 nm in diameter) act as paracrine mediators between MSCs and target cells.^{81,82} MSC-derived exosomes recapitulate the biological activity of MSCs and may serve as an alternative to whole cell therapy.^{83,84}

MSC-derived exosomes are enriched with therapeutic miRNAs, mRNAs, cytokines, lipids, mitochondria, and growth factors and represent a novel cell-free therapy with compelling advantages over parent MSCs as there is no risk of tumor formation and lower immunogenicity.⁸⁵⁻⁸⁷ Numerous *in vivo* studies have demonstrated that the therapeutic benefit of MSCs is principally orchestrated by the paracrine secretion of a broad repertoire of growth factors, chemokines, and cytokines.⁸⁸⁻⁹⁰ MSC-derived exosomes can mimic the effect of MSCs in various experimental models by inhibiting apoptosis and stimulating cell proliferation.⁹¹ It is known that exosomes may either transfer transcripts from injured cells to stem cells, resulting in reprogramming of their phenotype to acquire specific features of the tissue, or conversely, transcripts could be transferred from exosomes to injured cells, restraining tissue injury and inducing cell cycle re-entry of resident cells, leading to tissue self-repair.^{91,92} The underlying beneficial mechanistic pathways of MSC-derived exosomes involve the reprogramming of injured cells, cell proliferation and angiogenesis, and inhibition of cell apoptosis and inflammation.^{93,94} Several studies in animal models of AKI and CKD have demonstrated a potential of MSC-derived exosomes for preservation of renal structure and function, and these studies suggest that exosomes contribute to renal repair by virtue of their unique gene, microRNA, and protein cargo, which possess potent proregenerative properties, although more experimental and clinical studies are needed to further explore the mechanisms of MSC-derived exosomes' renoprotection.⁹⁵ Convincing evidence demonstrates the regenerative potential of exosomes released by stem cells, such as MSCs, in multiple kidney injury models. MSC-derived exosomes as a promising cell-free therapy in CKD have been well reviewed in a previous publication.⁹⁶ Thus MSC-derived exosomes may contribute to the recovery from renal injury. They are highly likely to have a therapeutic role in tissue regeneration after AKI and have the potential to transfer endogenous and exogenous therapeutic substances to recipient cells.⁹⁷⁻¹⁰¹

BONE MARROW-MESENCHYMAL STEM CELL-DERIVED-EXOSOMES IN KIDNEY REPAIR

BM-MSCs promote renal regeneration through ECVs such as BM-MSC-derived exosomes. It has been shown that MSC-derived exosomes significantly reduce their intrinsic regenerative potential in a mouse model of glycerol-induced AKI.^{102,103} BM-MSCs exert a renoprotective role through their paracrine antiinflammatory and antifibrotic properties.¹⁰⁴ BM-MSCs recruited to the site of injury may reprogram the damaged cells, stimulating their proliferation, and favoring tissue regeneration through the contribution of growth factors and transfer of genetic material by exosomes locally released by BM-MSCs.¹⁰⁵ The effect of these exosomes on the recovery of AKI was similar to that of human MSCs.¹⁰⁶ MSC-derived exosomes were reported to protect kidneys from IRI-induced AKI in rats by inhibiting apoptosis and stimulating tubular epithelial cell proliferation. Consequently, exosomes significantly improved the impairment of renal function and reduced transition of AKI to CKD.¹⁰⁷

Recent studies have demonstrated crucial roles of exosomes and tunneling nanotubes (TNTs) in long-range

intercellular transfer of cellular materials.¹⁰⁸ TNTs have been accepted as an emerging route of long-range intercellular communication, and cells can exchange cargo and organelles through TNTs. It has been reported that enhanced mitochondrial transfer between human corneal epithelial cells through TNTs promotes the regeneration of corneal epithelial cells in dry eye disease.¹⁰⁹ Microglia exposed to α -synuclein established a cellular network through TNTs, and lowering the α -synuclein burden attenuated the inflammatory profile of microglia and improved their survival.¹¹⁰ As reviewed and summarized, TNTs mediate intercellular trafficking of cellular cargo and organelles.^{111,112} The synergistic interplay between exosomes and TNTs in long-range intercellular transfer could be a critical repair mechanism in kidney injury.

ADIPOSE STEM CELL-DERIVED EXOSOMES IN KIDNEY REPAIR

The role of exosomes in renal inflammation has been examined in a porcine model of metabolic syndrome and renal artery stenosis, which was established in swine with unilateral renovascular disease complicated by metabolic syndrome (MetS).¹¹³ A single intrarenal delivery of exosomes harvested from autologous adipose tissue-derived MSCs was performed after 16 weeks of diet-induced MetS and renovascular disease, 4 weeks after exosome delivery. Stenotic-kidney renal blood flow and glomerular filtration rate (GFR) were measured *in vivo* (fast computed tomography), while characteristics of exosomes, renal microvascular architecture (micro-computed tomography), and injury pathways were analyzed *ex vivo*. The exosome fragments colocalized with stenotic-kidney tubular cells and macrophages and exosomes delivery attenuated renal inflammation and improved medullary oxygenation and fibrosis. The fall of renal blood flow and GFR in pigs with metabolic syndrome and renal artery stenosis was restored by exosome injection.¹¹³ This research group has further demonstrated that adipose tissue MSC-derived exosomes elicit better preservation of the stenotic-kidney microvasculature and greater attenuation of renal injury and fibrosis compared to percutaneous transluminal renal angioplasty (PTRA), possibly partly attributed to their cargo of vasculoprotective genes.⁹⁹ It has been reported that adipose tissue MSC-derived exosomes protected renal function in mice with AKI induced by cecal ligation and puncture (CLP) through the sirtuin 1 (SIRT1) pathway.¹¹⁴ Adipose MSC-derived exosomes protect against LPS-induced AKI more significantly compared to BM-MSC-derived exosomes and improved kidney function and structure by inhibiting oxidative stress and inflammation.¹¹⁵

ENDOTHELIAL PROGENITOR CELL-DERIVED EXOSOMES IN KIDNEY REPAIR

Endothelial progenitor cell-derived exosomes have been reported to prevent AKI in a rat model of IRI by enhancing tubular cell proliferation, reducing apoptosis, and leukocyte infiltration. These exosomes also protected against progression of chronic kidney damage by inhibiting capillary rarefaction, glomerulosclerosis, and tubulointerstitial fibrosis.¹¹⁶ After injection in rats with Thy1.1 glomerulonephritis, exosomes derived from EPCs localized within injured glomeruli and inhibited mesangial

cell activation, leucocyte infiltration, and apoptosis; increased serum complement hemolytic activity (CH50); decreased proteinuria; and ameliorated renal dysfunction, which suggest these exosomes exerted a protective effect by inhibition of antibody- and complement-mediated injury of mesangial cells.¹¹⁷

IPSC-DERIVED EXOSOMES IN KIDNEY REPAIR

iPSC-derived exosomes reduce cell death and inflammatory responses, protect functional mitochondria, and regulate several genes associated with oxidative stress in renal tubule cells with hypoxia-reoxygenation injury and in a rat model of AKI induced by IRI.¹⁰³ It was reported recently that iPSC-derived exosomes enhance the survival of HK-2 cells undergoing cisplatin-induced cell death in cell culture. In mice with cisplatin-induced AKI, iPSC-derived exosomes reduce renal functional impairment as shown by reduced blood urea nitrogen and creatinine, in association with decreased infiltration of immune cells, reduced expression of inflammatory genes in M1-induced THP-1 cells, and enhanced capillary density in the kidney of AKI mice. These exosomes suppressed inflammatory responses, inhibited tubular cell apoptosis, and reversed the activation of extracellular signal-regulated kinase 1/2 signaling in the kidney of mice with AKI.¹¹⁸ Another study demonstrated that hiPSC exosomes protect against renal injury in AKI induced by IRI via delivering SP1 into target renal cells and intracellularly, activating the expression of sphingosine kinase 1 and generation of sphingosine-1-phosphate (S1P).¹¹⁹

MESENCHYMAL STEM CELL-DERIVED EXOSOMES IN DIABETIC KIDNEY DISEASE

The renoprotective role of MSC-derived exosomes in DKD has been widely studied.^{120,121} It was reported that MSC exosomes attenuate diabetic nephropathy in a rat model of streptozotocin-induced diabetes mellitus through enhancement of autophagy activity.¹²² It was also reported that MSC-derived exosomes could attenuate DKD by decreasing cell apoptosis and inhibiting epithelial-to-mesenchymal transition (EMT) in diabetic kidneys in db/db mice. In vitro studies showed MSC-derived exosomes could reverse high glucose-induced apoptosis and EMT in HK2 cells, and miR-424-5p derived from MSC-derived exosomes inhibited YAP1 activation in HK2 cells, resulting in alleviation of high glucose-induced cell apoptosis and EMT.¹²³ Recent reports have demonstrated that MSC-derived exosomes attenuate inflammation including the expression of IL-6, IL-1 β , IL-18, and TNF- α ; depress the activation of the NLRP3 signaling pathway in podocytes under high glucose conditions and in diabetic mice; and ameliorate kidney injury. Knocking down miR-22-3p from MSC exosomes abolished their antiinflammation activity and beneficial function both in vitro and in vivo. These results indicate that MSC exosomes transferring miR-22-3p protected the podocytes and diabetic mice from inflammation, including by targeting the NLRP3 inflammasome.^{120,121} Adipose-derived stem cell exosomes regulate Nrf2/Keap1 in diabetic nephropathy by targeting FAM129B. ADSC-derived exosomes reversed oxidative stress and inflammation in podocytes and kidney tissues of diabetic mice induced by high glucose levels in vitro and in vivo, and FAM129B siRNA

blocked the inhibitory effect of ADSC-derived exosomes on intracellular reactive oxygen species and malondialdehyde upregulation induced by high glucose in podocytes.¹²⁴ The renoprotective role of MSC-derived exosomes has been well reviewed elsewhere.^{120,125} As summarized in these review articles, MSC-derived exosomes reduce ECM accumulation, thus improving fibrosis while protecting podocytes in models of DKD. However, the exosomes produced by various kidney resident cells are both crucial signaling mediators and potential biomarkers of DKD. These kidney cell exosomes facilitate cell-cell crosstalk as a contributing factor in DKD. Moreover, exosomes can be detected in urine and blood and have emerged as promising noninvasive biomarkers for the progress of DKD.

MESENCHYMAL STEM CELL-DERIVED EXOSOMES IN KIDNEY TRANSPLANTATION

After kidney transplantation, rats treated with MSC-derived exosomes showed improved renal function and survival. MSC-derived exosomes mitigated renal cell apoptosis, enhanced proliferation, alleviated inflammation at the early stage, and reduced renal fibrosis at the late stage of transplantation. MSC-derived exosomes also decreased the number of CD68⁺ macrophages and expression levels of α -SMA and TGF- β 1 proteins while increasing the protein expression level of HGF in the kidney.¹²⁶ MSC-derived exosomes also prevented ischemic injury in isolated rat kidneys, as MSC-derived exosomes upregulate the expression of three genes encoding enzymes known to improve cell energy metabolism and three genes encoding proteins involved in ion membrane transport. It is suggested that MSCs-derived exosomes protect the kidney from ischemic injury by preserving the enzymatic machinery essential for cell viability and protecting the kidney from reperfusion injury.¹²⁷

KIDNEY ORGANOIDS

INDUCED PLURIPOTENT STEM CELL-DERIVED KIDNEY ORGANOIDS

In 2015, an efficient, chemically defined protocol for differentiating hPSCs into multipotent nephron progenitor cells (NPCs) that can form nephron-like structures was reported.¹²⁸ In 3D culture, NPCs form kidney organoids containing epithelial nephron-like structures expressing markers of podocytes, proximal tubules, loops of Henle, and distal tubules in an organized, continuous arrangement that resembles the nephron in vivo.¹²⁸ In the same year, kidney organoids that contained nephrons associated with a collecting duct network surrounded by renal interstitium and ECs were generated.¹²⁹ The distal and proximal tubules, early loops of Henle, and glomeruli containing podocytes elaborating foot processes and undergoing vascularization were present in the nephrons of these organoids. The transcription profiles of these kidney organoids showed highest congruence with the first trimester of human fetal kidney. It is well accepted that dysfunction of podocytes and proximal tubular cells mainly contribute to glomerulosclerosis and tubular interstitial fibrosis, respectively, in CKD. Thus iPSC-derived kidney organoids have emerged as advanced in vitro models of kidney development, physiology, and disease. Current

conventional research models, such as static cell cultures and animal models, are insufficient to reflect the complex human *in vivo* situation or lack translational value. Kidney organoids resemble the human kidney in a cell culture dish. Thus they can be applied in regenerative medicine, human kidney disease models, and drug development (“clinical trials in a dish”) and greatly reduce the usage of animals.¹²⁹

In comparison with immortalized 2-dimensional human podocyte cell lines, 3D human glomeruli sieved from iPSC-derived kidney organoids show improved podocyte-specific gene expression and an improved glomerular basement membrane matrisome. Hence human iPSC-derived organoid glomeruli represent an accessible approach to the *in vitro* modeling of human podocytopathies and screening for podocyte toxicity.¹³⁰ The applications of kidney organoids are currently limited due to organoid variability, nephron immaturity, low throughput, and limited scale. Extrusion-based 3D cellular bioprinting to deliver rapid and high-throughput generation of kidney organoids with highly reproducible cell number and viability has been tested. The automated extrusion-based bioprinting for kidney organoid production delivers improvements in throughput, quality control, scale, and structure, facilitating *in vitro* and *in vivo* applications of stem cell-derived human kidney tissue.¹³¹ Kidney organoids that transcriptomically match second-trimester human fetal kidneys have been generated in an *in vivo* microenvironment that promotes the growth and differentiation of implanted kidney organoids, as well as providing a vascular component, which demonstrates that soft hydrogel accelerates the differentiation of hPSC-derived kidney organoids.¹³²

ADULT STEM CELL-DERIVED KIDNEY ORGANOID

Primary kidney tubular epithelial organoids have been established from human and mouse kidney tissue and can be expanded for at least 20 passages while retaining a normal number of chromosomes. These organoids can also be established from human urine-derived stem cells (USCs).¹³³ These human organoids represent proximal and distal nephron segments and have been used to model infectious, malignant, and hereditary kidney diseases in a personalized fashion.¹³³ A 3D *in vitro* human renal organoid model has been developed from adult differentiated cells without gene modification. It is composed of multiple cell types and has been evaluated for morphology, viability, and erythropoietin production. This renal organoid is proposed as a model for drug screening and nephrotoxicity assays.¹³⁴ The generation of an expandable, 3D branching ureteric bud (UB) organoid culture model that can be derived from primary UB progenitors from mouse and human fetal kidneys or generated *de novo* from human pluripotent stem cells has also been reported. Aggregating 3D-cultured NPCs with UB organoids *in vitro* results in a reiterative process of branching morphogenesis and nephron induction, similar to kidney development.¹³⁵ Thus autologous ASC-derived organoids from patients may prove a useful tool for precision medicine.

The reduced survival time and small size of kidney organoids have led to challenges that have limited its capacity to study renal regeneration. It has been reported that kidney progenitors expressing nephrons develop from human pluripotent stem cells (hPSCs) subcutaneously

implanted into immunodeficient mice. After 12 weeks, implants formed organ-like masses, which included perfused glomeruli containing human capillaries, podocytes with regions of mature basement membrane, and mesangial cells. After intravenous injection of fluorescent dextran, signal was detected in tubules, demonstrating uptake from glomerular filtrate. The functioning nephrons formed from hPSC-derived kidney precursors *in vivo* have been developed to model and treat kidney diseases.¹³⁶ Similarly, vascularized 3D kidney organoids have been generated using dynamic modulation of WNT signaling to control the relative proportion of proximal versus distal nephron segments, producing a correlative level of vascular endothelial growth factor A (VEGFA) to define a resident vascular network. These kidney organoids undergo further structural and functional maturation upon implantation.¹³⁷

Despite great efforts from international researchers and rapid development and advancements in kidney organoid generation, there are challenges and limitations in clinical applications.¹³⁸ For example, kidney organoids lack some critical cell types (e.g., infiltrating immune cells) and structures (e.g., blood vessels), lack of maturity, lack of ongoing nephron formation, and lack of systemic hemodynamic-related influences, as well as exposure to complex metabolic milieus in diabetic and other metabolic diseases. Another challenge lies in reproducibility due to technical variations in the study of kidney disease using kidney organoids. However, the robustness of the protocols and simplicity of organoid generation provide great benefits by using human/patient-based approaches to study kidney disease compared with 2-dimensional immortalized or primary cell line approaches. ASC-derived organoids to date have been more useful to define therapies for kidney-specific problems, such as autosomal dominant polycystic kidney disease (ADPKD). Hence modeling for personalized therapies to date has largely included ADPKD compared with glomerular and metabolic disease, for which systemic immune/inflammatory/metabolic mechanisms are more at play.

DE NOVO WHOLE KIDNEY DEVELOPMENT

De novo whole kidney development is defined as functional whole kidney regeneration from stem cells, which is the “holy grail” for patients with CKD.

KIDNEY ORGANOID AS A PRECURSOR OF WHOLE KIDNEY DEVELOPMENT

Regenerative medicine aims to replace or regenerate damaged cells, tissues, and ultimately the kidney to completely restore normal renal structure and function. As described earlier, kidney organoids that contain nephrons associated with a collecting duct network surrounded by renal interstitium and ECs have been generated,¹²⁹ but significant limitations exist. Despite various strategies to establish functional kidney organoids, the therapeutic benefit of organoids before translation to clinical use warrants many further studies. As well summarized by Vives and Battle-Morera: “The efficacy and safety of organoid-based therapies will require appropriate cellular composition, proper engraftment and vascularization into the host, and the demonstration of functional activity. Translation of

organoid technologies into the clinic may benefit from the emergence of human leukocyte antigen (HLA)-homozygous iPSC initiatives, which hold the potential to foster the development of patient-compatible regenerative therapeutic approaches, and the blending of organoid technology with 3D bioprinting and vascularization approaches may lead to macrostructures with the desired cellular composition for successful transplantation.”¹³⁹

3D BIOPRINTING BASED ON ENGINEERING TECHNOLOGY

3D bioprinting has the potential to be used to “rebuild” a kidney. Bioprinting is a technology to allow the precise deposition of biomaterial inks layer by layer to fabricate complex tissues or organ models.¹⁴⁰ Cellular extrusion-based 3D bioprinting has been used to deliver rapid and high-throughput generation of kidney organoids with highly reproducible cell number and viability. 3D bioprinting enables precise manipulation of biophysical properties including organoid size, cell number, and conformation, with modification of organoid conformation substantially increasing nephron yield per starting cell number, which facilitates the manufacture of uniformly patterned kidney tissue sheets with functional proximal tubular segments.¹³¹ Using Organovo’s proprietary 3D bioprinting platform, King and colleagues¹⁴¹ developed a fully cellular human in vitro model of the proximal tubule interstitial interface comprising renal fibroblasts, ECs and primary human renal proximal tubule epithelial cells to enable a more accurate prediction of tissue-level clinical outcomes. Histologically, the extensive microvascular networks are supported by endogenous extracellular matrix deposition, the epithelial cells of the 3D proximal tubule tissues demonstrated tight junction formation and expression of renal uptake and efflux transporters, and the polarized localization and function of P-glycoprotein (P-gp) and sodium-glucose transport protein 2 (SGLT2) were confirmed. The tissues also demonstrated nephrotoxicity of cisplatin and a fibrotic response to TGFβ.¹⁴¹ This research has successfully opened a door to kidney regeneration and organ transplantation. However, the challenge for 3D bioprinting of kidneys is to develop “renal” printing ink that will resemble complex renal structures, ECM composition, and both glomerular and tubular functionality.¹⁴² Despite advances in 3D bioprinting technology, there are still limitations in clinical applications, such as the ability of cells to survive in the printing process, the print speed, temperature, and pressure, all of which will need to be optimized for different cell types. Additionally, once developed, these complex 3D tissues can only be maintained for a relatively short period.^{143,144}

DECELLULARIZED KIDNEYS

Kidney decellularization is an alternative strategy to develop a kidney scaffold, which can then theoretically be used to “rebuild” a kidney. Briefly, all cells are removed throughout the whole kidney from the host (autologous kidney) with detergent, and then various sources of cells such as stem cells or adult differentiated cells from the host are injected to differentiate into mature cells in the decellularized scaffold with the extracellular matrix (ECM) to form a recellularized kidney, with functionality and no immune response after in vivo transplantation on the host. Ross and colleagues¹⁴⁵

decellularized intact rat kidneys in a manner that preserved the intricate architecture and seeded them with pluripotent murine embryonic stem cells (mESCs). Similar studies reported that the decellularized intact rat kidney scaffolds preserve the 3D architecture of blood vessels, glomeruli, and tubuli, and histologically, collagen IV, laminin, and fibronectin staining of decellularized scaffolds were similar to those of native kidney tissues.¹⁴⁶ Urine production from the recellularized kidneys was reported.¹⁴⁷ Additionally, kidneys were recellularized by ex vivo infusion of hiPSC-ECs through the renal artery and vein of acellular kidneys, which resulted in the uniform distribution of the cells in all the vasculature compartments, from glomerular capillaries to peritubular capillaries and small vessels, and the presence of continuously distributed cells along the vessel wall was evidenced.¹⁴⁸ The biomimetic scaffold for kidney regeneration was assessed in a rat kidney cortical defect model.¹⁴⁹ Despite early promise, this strategy has not been translated into viable options for organ regeneration due to the complexity of kidney structure and influences of systemic factors.

CLINICAL STUDIES OF STEM CELLS IN ACUTE KIDNEY INJURY AND CHRONIC KIDNEY DISEASE

To date, there is no FDA-approved stem cell-based treatment for kidney disease. However, increasing evidence in preclinical studies, demonstrating the therapeutic role of stem cells in kidney injury, has prompted clinical studies to investigate the role of stem cells in patients with either acute or chronic kidney injury.

STEM CELLS IN ACUTE KIDNEY INJURY

The safety and efficacy of AC607 for the treatment of kidney injury in patients undergoing cardiac surgery have been examined (NCT01602328). AC607s are expanded BM-MSCs, obtained from healthy adult donors using a proprietary manufacturing process. Patients who experienced kidney injury within 48 hours of their cardiac surgery were enrolled in the study. Once enrolled, subjects received a single administration of AC607 or placebo. Kidney recovery was evaluated over the subsequent 30 days and death, or the need for dialysis was evaluated within 90 days of dosing. At follow-up, 12 patients who received AC607 and 6 patients who received placebo had died. Rates of other adverse events did not differ between groups. This trial was terminated prematurely due to futility because after 156 patients had been enrolled, time to renal function recovery, the need for dialysis, and 30-day all-cause mortality were not significantly different in the MSC- compared with placebo-treated group.¹⁵⁰

A further phase 1 clinical trial (NCT00733876) was undertaken to assess the safety of administering allogenic MSCs in patients at high risk of developing AKI after undergoing on-pump cardiac surgery. This trial was completed in October 2013. The results suggested that MSCs protected against early and late postsurgical development of AKI reduced the length of hospital stay and need for patient readmissions in the absence of MSC-specific adverse or serious adverse events.¹⁵¹

A multicenter, randomized, sham-controlled, double-blind, ascending-dose study of extracorporeal MSCs (SBI-101 Therapy) in subjects with AKI receiving continuous kidney replacement therapy (NCT03015623) was undertaken more recently. The results indicate that treatment with SBI-101 elicits an immunotherapeutic response that triggers an accelerated phenotypic switch from tissue injury to tissue repair.¹⁵²

NCT01275612 was a pilot, explorative study to test the feasibility and safety of systemic infusion of donor ex vivo expanded MSCs to repair the kidney and improve function in patients with solid organ cancers who develop acute renal failure after chemotherapy with cisplatin. This phase 1 clinical study was withdrawn without results concluding on March 19, 2018.

STEM CELLS IN CHRONIC KIDNEY DISEASE

Clinical trial NCT02166489 was a phase 1 clinical trial with 12-month follow-up in 6 patients with ADPKD, completed in January 2016. This study was designed to provide confirmation of safety of MSC therapy in chronic renal failure due to ADPKD. This trial demonstrated that an intravenous infusion of autologous MSCs was safe and well tolerated in ADPKD patients. However, the efficacy of BM-MSCs was not able to be assessed due to the trial design, and a randomized placebo-controlled trial with a larger population was recommended.¹⁵³

In NCT02195323, the 18-month safety and potential efficacy of autologous MSCs as a therapy for CKD was assessed. This study was completed in 2016. Safety and tolerability were demonstrated in seven CKD patients with a single-dose infusion of autologous BM-MSCs. No cell-related adverse events were observed during the trial.¹⁵⁴

The phase I clinical trial NCT02266394 entitled “Hypoxia and inflammatory injury in human renovascular hypertension: phase 1 trial of MSCs therapy” investigated autologous AD-MSCs in atherosclerotic renovascular disease. The trial compared the administration of AD-MSCs to standard medical therapy alone, with patients matched for age, kidney function, and blood pressure. A total of 21 patients were treated with different dose levels (1, 2.5, and 5.0×10^5 cells/kg) with 7 patients in each dosing group. The results showed that mean renal blood flow in the stenotic kidney significantly increased in the entire cohort of stem cell infusion-treated subjects compared with the baseline (164–190 mL/min). Mean systolic blood pressure in the stem cell-treated cohort fell significantly (144–136 mm Hg) compared with baseline and the mean GFR increased modestly but significantly from 53 to 56 mL/min. Hypoxia, renal vein inflammatory cytokines, and angiogenic biomarkers decreased significantly following stem cell infusion.¹⁵⁵

The safety, tolerability, and efficacy of a single intravenous infusion of two doses of mesenchymal precursor cells versus placebo in subjects with DKD have been assessed (NCT01843387). This study demonstrated the safety of adult allogeneic bone marrow-derived mesenchymal precursor cells (MPCs) (rexlemestocel-L) in DKD with suggestive beneficial effects on renal function to be confirmed in larger, appropriately powered trials.¹⁵⁶

STEM CELLS IN KIDNEY TRANSPLANTATION

The effect of MSCs on the T cell repertoire of kidney transplant patients has been investigated (NCT02409940). The results suggest that autologous MSCs can be used safely in patients undergoing living donor renal transplantation, leading to expansion of regulatory T cells and a decrease in T cell proliferation. Larger randomized trial studies are needed to confirm these findings and evaluate whether this will have any impact on immunosuppressive therapy.¹⁵⁷

Autologous MSCs as an alternative for antibody induction therapy in renal transplantation were assessed in NCT00658073. Use of autologous MSCs compared with anti-IL-2 receptor antibody induction therapy resulted in a lower incidence of acute rejection, decreased risk of opportunistic infection, and better estimated renal function at 1 year.¹⁵⁸

As shown in Table 84.2, there are ongoing registered clinical trials to investigate the role of stem cells in CKD.

GENE AND CELL THERAPY OTHER THAN STEM CELL THERAPY

In addition to stem cells, additional cells such as lymphocytes, dendritic cells, and pancreatic islet cells can be used to modify diseases, including kidney disease. Gene therapy can treat certain genetic or inherited disorders by replacing damaged genes, turning off problematic genes, or inserting new genes to normalize the gene/protein environment required for health. Cell-based gene therapy is a combination of cell therapy and gene therapy (e.g., genetically modified cells are infused to treat specific genetic disorders).

Peek and Wilson have described the potential for cell and gene therapy in the treatment of kidney disease in a review that focused on recent genetic studies, key advances, and emerging technologies. Several crucial considerations for renal genetic and cell therapies were described.¹⁵⁹ Davis and Park described a comprehensive overview of the most common replication-defective viral vector systems based on either adenovirus, lentivirus, or adeno-associated virus for genetic modification of various cell types in the kidney over the past quarter century.¹⁶⁰ Nanoparticles and nonviral delivery for gene therapy in kidney disease have also been reported. The role of renal proximal tubular NF- κ B essential modulator (NEMO) in ischemic AKI was examined in mice through targeted renal proximal tubular NEMO inhibition with mesoscale nanoparticle-encapsulated NEMO binding peptide. The results showed nanoparticle-encapsulated NEMO binding peptides protected against ischemic AKI through decreased renal tubular necrosis, inflammation, and apoptosis compared with the control.¹⁶¹ However, it is now proposed that exosomes derived from stem cells may be more appropriate to deliver targeted gene therapy.

The role of a nonviral tubular-targeted system, KIM-1 specific EVs in AKI has been reported. Through these EVs, P65 and Snail siRNA are targeted to the kidney. Dual suppression of P65 and Snail by siRNAs significantly improved I/R- and UUO-induced kidney injury by alleviating tubulointerstitial inflammation and fibrosis and potentially abrogated the transition to CKD.¹⁶²

The increasing identification of pathogenic DNA sequence variants with monogenic high penetrance or

Table 84.2 Currently Ongoing Clinical Trials of Stem Cell Therapy in Chronic Kidney Disease

Trial ID	Condition	Title	Link	Location
NCT04125329	DKD	Umbilical Cord Mesenchymal Stem Cells Therapy for Diabetic Nephropathy	https://clinicaltrials.gov/study/NCT04125329?term=stem%20cell&cond=Kidney%20Diseases&aggFilters=status:rec&city=&rank=1	Kunming, China
NCT02176434	ESKD	Pilot Feasibility Study of Combined Kidney and Hematopoietic Stem Cell Transplantation to Cure End-stage Renal Disease	https://clinicaltrials.gov/study/NCT02176434?term=stem%20cell&cond=Kidney%20Diseases&aggFilters=status:rec&city=&rank=2	Zurich, Switzerland
NCT04869761	CKD	Stem Cell Therapy for Chronic Kidney Disease	https://clinicaltrials.gov/study/NCT04869761?term=stem%20cell&cond=Kidney%20Diseases&aggFilters=status:rec&city=&rank=3	Rochester, Minnesota, USA
NCT05512988		Human Umbilical Cord Mesenchymal Stem Cell (UC-MSC) Delayed Renal Chronic Kidney Disease (CKD34)	https://clinicaltrials.gov/study/NCT05512988?term=stem%20cell&cond=Kidney%20Diseases&aggFilters=status:rec&city=&rank=4	Wuhan, China
NCT05018845	CKD	Safety of Cultured Allogeneic Adult Umbilical Cord-Derived Mesenchymal Stem Cell Intravenous Infusion for CKD	https://clinicaltrials.gov/study/NCT05018845?term=stem%20cell&cond=Kidney%20Diseases&aggFilters=status:rec&city=&rank=6	St. John's, Antigua and Barbuda
NCT03939741	CKD	SVF (Adipose Tissue Derived MSC) Based Therapy for CKD	https://clinicaltrials.gov/study/NCT03939741?term=stem%20cell&cond=Kidney%20Diseases&aggFilters=status:rec&city=&rank=7	Dhaka, Bangladesh
NCT04392206	ESKD	AMSC for Reducing Anastomotic Stenosis in Primary Arteriovenous Anastomoses	https://clinicaltrials.gov/study/NCT04392206?term=stem%20cell&cond=Kidney%20Diseases&aggFilters=status:rec&city=&rank=9	Jacksonville, Florida, USA
NCT03673748	Lupus nephritis	Treatment of Lupus Nephritis With Allogeneic Mesenchymal Stem Cells	https://clinicaltrials.gov/study/NCT03673748?term=stem%20cell&cond=Kidney%20Diseases&aggFilters=status:rec&city=&rank=12	Valladolid, Spain
NCT05508009	CKD	Early Trial of Allogeneic Hematopoietic Stem Cell Transplantation for Patients Who Will Receive a Kidney Transplant from the Same Donor	https://clinicaltrials.gov/study/NCT05508009?term=stem%20cell&cond=Kidney%20Diseases&aggFilters=status:rec&city=&rank=19	Palo Alto, California, USA
NCT05631717	Lupus nephritis	The Study of Comparing the Efficacy and Safety of Human Umbilical Cord MSCs and Low-dose IL-2 in the Treatment of LN	https://clinicaltrials.gov/study/NCT05631717?term=stem%20cell&cond=Kidney%20Diseases&aggFilters=status:rec&city=&rank=21	Nanjing, China
NCT06058078	Lupus nephritis	RY_SW01 Cell Injection Therapy in Active Lupus Nephritis	https://clinicaltrials.gov/study/NCT06058078?term=stem%20cell&cond=Kidney%20Diseases&aggFilters=status:rec&city=&rank=22	Nanjing, China

partial penetrance inheritance patterns, in the etiology of many forms of CKD, has given rise to the emergence of genomically precise therapies. Several of these involve RNA interference (RNAi) and antisense oligonucleotide technologies. A prominent example is the targeting of hepatic enzymes glycolate oxidase with lumasiran and lactate dehydrogenase A (LDH-A) with nedosiran in relevant forms of primary hyperoxaluria. RNAi inhibition of an upstream step in the production of oxalate results in a significant reduction of urinary oxalate excretion in PH1 patients, with ongoing monitoring of safety and of decreased oxalate deposition and kidney stones and preservation of kidney function.¹⁶³ This potential transformative approach may preempt the need for liver and kidney transplantation in PH1 patients and families. Another example involves ongoing clinical trials using antisense oligonucleotides to inhibit production of the APOL1 gene product in subjects with high-risk genotypes and at this locus together with evidence of kidney injury. More detailed information about genomically precise therapies and gene therapies is provided in Chapter 44.

LIMITATIONS OF STEM CELL THERAPY AND FUTURE DIRECTIONS

To date, most clinical trials have shown the safety and tolerability of various types of stem cell therapies in short-term studies in humans. Many preclinical studies have demonstrated the renoprotective role of stem cell therapy in AKI and CKD. Successful outcomes from those animal studies have not been reproduced to date in human clinical trials. The variations between animal and human physiology may have led to the failure to translate results of positive renoprotection of stem cells from preclinical studies to a clinical application. Both AKI and CKD in experimental animals are generally induced by a single insult, such as diabetes, a cytotoxic reagent/drug, or ischemia, which are artificial disease models. However, the development of AKI and CKD in humans involves complex mechanisms. Thus it is difficult to precisely reproduce the results of preclinical studies in human studies. Kidney organoids are in vitro kidney models that are generated from stem cells in 3D cultures. Kidney organoids established from human MSCs and their production of exosomes may provide an excellent platform for kidney disease modeling to study the therapeutic role of stem cells.

There are challenges in the clinical translation of stem cell therapeutics. ESCs have limited clinical application due to ethical issues regarding their origin, tumorigenicity after transplantation. The clinical efficacy of MSCs is limited due to low engraftment, poor survival rate, and impaired paracrine capacity in vivo. Although iPSCs are highly proliferative in vivo, the use of viral vectors for reprogramming leads to potential risk of tumor development after transplantation. As described earlier, stem cell-derived exosomes should be considered as a possible alternative to live stem cells in kidney disease.

Collectively, as the limited results available in clinical trials still lack long-term follow-up, it is necessary to perform appropriately designed clinical studies to define long-term safety of these stem cell-based therapies in patients with

(or at risk of) AKI, CKD, and kidney transplantation. The potential risks of fibrosis, abnormal differentiation, or malignancy induced by stem cells in vivo must be rigorously examined before widespread clinical use is endorsed. Stem cell-based exosome and cell-based gene therapeutics will potentially have a place in the therapy of kidney disease if renoprotective mechanisms are comprehensively understood and the source of cells, cell dose, route, and timing of administration, and repeated dosing is defined. Longer-term safety and toxicity studies are also needed. To date, renal regenerative strategies remain in their infancy for clinical use but are useful for disease modeling and testing for nephrotoxicity of novel therapeutics.

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